Blockchain-Based Certificateless Conditional Anonymous Authentication for IIoT

Xinchao Wang, Wei Wang, *Member, IEEE*, Cheng Huang, *Member, IEEE*, Ping Cao, Youwen Zhu, and Qihui Wu, *Senior Member, IEEE*

***Abstract*—Identity authentication is an essential element for industrial Internet of Things (IIoT), which guarantees secure access control for various devices. Existing authentication schemes face some security threats, including temporary secret leakage attack, key recovery attack, and forgery attack. In this article, we introduce a blockchain-based certificateless conditional anonymous authentication (BCCA) scheme specifically designed for IIoT. To optimize the authentication efficiency, BCCA employs elliptic curve design to avoid the relatively time-consuming bilinear pairing operation. Additionally, we introduce a precomputation strategy, allowing users to prepare essential materials in advance, and one-time verification support for batch signatures is applied, thus reducing authentication latency. To further enhance the security, random verification checksums are employed to counter key recovery attack, and a combination of long-term and short-term secrets is used to mitigate temporary secret leakage attack. Simulation results demonstrate that our scheme has advantages in both security and computational cost.**

***Index Terms*—Authentication, blockchain, certificateless, industrial Internet of Things (IIoT), security.**

Nomenclature

Notation	Description
U_i	ith user.
RID_i	Real identity of U_i.
VS_i	ith consensus server.
P_{pub}	System public key.
s	System master key.
ID_i	Anonymous identity of U_i.
sk_i	Full secret key of U_i, where $\mathrm{sk}_i = (x_i, \mathrm{psk}_i)$.
pk_i	Full public key of U_i, where $\mathrm{pk}_i = (\mathrm{upk}_i, \mathrm{gpk}_i)$.
$\mathbb{G}$	Cyclic additive group $\mathbb{G}$ with order q.
P	Generator of $\mathbb{G}$.
m_i	Message collected from the ith user.
c_i	Ciphertext of m_i.
Enc	Symmetric encryption operation.
σ_i	Signature of the c_i.
Notation of Operation	
T_{Ha}	Time of map-to-point hash operation.
T_{Pa}	Time of bilinear pairing operation.
T_{Mu}	Time of elliptic curve scalar multiplication operation.
T_{Add}	Time of elliptic curve point addition operation.
T_{mu}	Time of modular multiplication operation.
T_{inv}	Time of modular inversion operation.
T_{ha}	Time of general hash operation.

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Xinchao Wang, Wei Wang, Ping Cao, and Qihui Wu are with the College of Electronic and Information Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China (e-mail: xinchao_wang@nuaa.edu.cn; wei_wang@nuaa.edu.cn; caoping@nuaa.edu.cn; wuqihui2014@sina.com).

Cheng Huang is with the Department of Electrical and Computer Engineering, University of Waterloo, Waterloo, ON N2L 3G1, Canada (e-mail: c225huan@uwaterloo.ca).

Youwen Zhu is with the College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, China (e-mail: zhuyw@nuaa.edu.cn).

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I. Introduction

THE industrial Internet of Things (IIoT) is a network of intelligent and highly connected industrial components. It enables real-time monitoring, efficient management, and control of industrial processes, assets, and operational time, resulting in increased productivity and reduced operational costs [1]. It is extensively applied across diverse domains, encompassing smart city [2], smart grid [3], and smart factory [4].

The explosive growth of heterogeneous devices in IIoT has led to a dramatic increase in the amount of data and potential vulnerabilities. Currently, IIoT data acquisition is primarily through sensors, and data must be transmitted via wireless networks. Due to the openness of the wireless network, data are always subject to various attacks during transmission [5], [6], such as forgery attack, man-in-the-middle attack, and replay attack. These attacks make it difficult to guarantee reliability and timeliness of data for decision analysis, resulting in disruption of normal network operations. Therefore, ensuring communication security in IIoT is crucial, requiring a reliable authentication mechanism to prevent unauthorized access.

Presently, some IIoT authentication systems employ a design approach based on public key infrastructure (PKI), but they suffer from the complexity of certificate management. To solve the certificate management problem, some IIoT systems have been designed using identity-based public key cryptography [7]. However, it has inherent key escrow weaknesses. Therefore, the above-mentioned two schemes may not meet the efficiency and secure requirements in IIoT. In order to address the aforementioned issues, we have employed a certificateless public key cryptosystem [8] in IIoT.

However, traditional certificateless schemes face several security and efficiency challenges. First, some schemes [9], [10],

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[11], [12] require computationally expensive bilinear pairing operations, resulting in lower authentication efficiency. Additionally, some schemes [13], [14] are vulnerable to key recovery attack, where adversaries can decrypt fixed private keys by collecting multiple signatures. Moreover, some schemes [15], [16] are susceptible to forgery attack, as adversaries can simulate signatures. Furthermore, many schemes [17], [18] do not consider the potential risk of temporary secret leakage in IIoT, where attackers can exploit factors obtained through power analysis or other means for malicious impersonation.

In addition, some certificateless authentication systems, relying on trusted third parties, suffer from single points of failure. To address this limitation, blockchain technology [19] has gained significant attention due to its decentralized, traceable, and tamper-resistant characteristics. By leveraging blockchain, IIoT can benefit from reliable supervision and data traceability. Through the collaboration of multiple mutually untrusted organizations, distributed ledgers can be maintained based on specific contractual constraints, effectively mitigating the risk of single points of failure.

While several blockchain-based certificateless authentication schemes [9], [16], [20] have been proposed, updating information on the blockchain still necessitates the creation of new blocks and achieving consensus throughout the entire blockchain network. Retrieving updated information also entails traversing the entire blockchain, incurring substantial computational and time costs, and imposing additional storage pressure on the blockchain system. Given that our scheme integrates blockchain technology and simultaneously demands real-time updates of on-chain data to meet the requirements of IIoT, as well as enhancing regulatory flexibility, such as the need for the removal of specific information from the blockchain, and considering the possibility of malicious reporting, timely correction of such information is required to protect the rights of those reported. Therefore, an efficient on-chain modifiable mechanism is required to meet the demands of IIoT.

In summary, the design of certificateless schemes necessitates a focus on improving both security and efficiency. Additionally, integrating chameleon hash (CH) technology becomes a crucial avenue for enhancing regulatory flexibility within the blockchain, making it more applicable to IIoT.

A. Contributions

To solve the aforementioned deficiencies, we design a blockchain-based certificateless conditional anonymous authentication (BCCA) scheme, namely BCAA, utilizing a dual-chain model. Our goal is to enhance system security while ensuring authentication efficiency. The primary contributions of this work can be outlined as follows.

1) We construct a dual-chain model for blockchain-based certificateless anonymous authentication. The model involves storing signature history and evidence separately on the historical chain and the evidence chain, respectively. By incorporating conditional identity anonymity techniques, our approach enables traceability of signatures and provides the ability to revoke malicious users. Notably, we have independently propose a CH algorithm to ensure the rectification of illegal information on the evidence chain.
2) We construct a secure and efficient certificateless signature scheme based on the schnorr signature mechanism. During the registration phase, a preprocessing strategy is employed, allowing users to prepare the required key materials for signature generation in advance, thereby enhancing the efficiency of signature generation. Additionally, our scheme supports batch signature verification, further improving the efficiency of signature authentication. Moreover, in the login phase for industrial controllers, we adopt a multifactor authentication approach combining smart cards, biometric data, and passwords, making it impossible for attackers to impersonate legitimate users and gain access to industrial controllers.
3) We construct a security analysis that shows the proposed scheme's security and resistance against various attacks (e.g., man-in-the-middle attack, replay attack, forgery attack). In the random oracle model, we prove that BCCA is secure against both Types I and II attacks. Furthermore, our performance evaluation demonstrates that BCCA is lightweight in computational cost.

B. Related Work

To establish a trusted connection in the IIoT, Verma et al. [21] presented the first nonpairing certificate-based proxy signature scheme. To reduce the bandwidth demand in IIoT, Verma et al. [22] further proposed the first nonpairing certificate-based aggregate signature scheme. Subsequently, many identity authentication schemes for IIoT have emerged [23], [24], [25], [26]. However, they all rely on PKI, which brings about complex certificate management.

In order to eliminate certificate management, some identity-based authentication schemes have been proposed in [27], [28], and [29]. Specifically, Karati et al. [27] proposed a new identity-based signcryption scheme for IIoT, which achieves higher efficiency and stronger security. Meanwhile, Li et al. [28] presented the first anonymous aggregate signature scheme for IIoT, which ensures data traceability and nonrepudiation using blockchain technology. However, although identity information is only known to users, it is vulnerable to security threats when exposed in smart contract for authentication. Additionally, Tan et al. [29] introduced a lightweight authentication scheme supported by blockchain technology for industrial unmanned aerial vehicles. However, due to the exclusive use of fixed private secrets in the authentication process, attackers can decrypt them by collecting multiple authentication information, thereby making the scheme susceptible to security threats. Subsequently, Wu et al. [30] proposed an ideal framework for future dynamic spectrum sharing and highlighted the vulnerability of centralized architectures to malicious attacks in crowd sensing. Furthermore, they pointed out that the spectrum sensing data submitted by participants may contain sensitive user information, which can potentially lead to privacy breaches. Huang et al. [31] proposed a blockchain-assisted personalized car insurance scheme that combines consortium blockchain, partially homomorphic encryption, and zero-knowledge proof techniques. It achieves public verification of data collection and processing, as well as privacy preservation in driving behavior evaluation.

Recently, Gayathri et al. [17] proposed a pairing-free certificateless aggregate signature scheme for medical wireless sensor networks. However, the scheme has an aggregation vulnerability where attackers can swap the positions of two signatures, resulting in successful verification but no nonrepudiation guarantee. Thumbur et al. [32] proposed an efficient pairing-free

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certificateless signature scheme specifically designed for securing communication in resource-constrained devices. However, Xu et al. [33] have discovered a vulnerability in the scheme proposed by Thumbur et al. [32], specifically related to signature forgery attacks. Mei et al. [15] proposed an efficient certificateless aggregate signature scheme with conditional privacy protection, but their scheme is also susceptible to signature forgery attack. Recently, Liu et al. [16] proposed a pairing-free certificateless aggregation signature scheme, although the scheme exhibits susceptibility to forgery attack. Xiang et al. [18] proposed a novel pairing-free certificateless signature scheme for Internet of Things (IoT). However, Ma et al. [34] found that the scheme proposed by Xiang et al. [18] is not resistant to signature forgery attack. Wang et al. [13] proposed a blockchain-based certificateless signature scheme designed for IIoT. However, the signature generation process only uses private keys without temporary random numbers, which may lead to the key-recovery attack when attackers collect enough signatures. Han et al. [14] presented a pairing-free certificateless aggregate signature scheme, but it also suffers from the key-recovery attack problem. Furthermore, the authors in [13] and [14] are also vulnerable to signature forgery attack.

Inspired by [35], we need to consider the scenario of random number leakage. We have noticed that many certificateless schemes typically use a random number as the signature's checksum without considering the potential key recovery attack that may result from random number leakage. Therefore, we propose combining random numbers with a user's long-term secrets, such as biometric factors, to construct the temporary checksums.

In conclusion, traditional certificateless schemes face several security and efficiency challenges. While some schemes have adopted elliptic curve cryptography (ECC)-based signature algorithms to construct more efficient authentication methods, there is still room for improvement. Existing authentication schemes are vulnerable to forgery attack, temporary secret leakage attack, key recovery attack, and single points of failure issue. Additionally, the integration of certificateless schemes with blockchain lacks a flexible on-chain modification mechanism. Therefore, it is imperative to propose an authentication scheme suitable for IIoT with high computational efficiency in order to enhance security and supervisory flexibility.

The rest of this article is organized as follows. Section II presents the preliminaries. The proposed BCCA scheme is presented in Section III. The security analysis of BCCA is provided in Section IV. Section V gives the performance evaluation of the scheme. Finally, Section VI concludes this article.

II. Preliminaries

In this section, we first introduce the computational hardness assumption as outlined in [13] and [18]. Next, we describe related theories, such as CH and fuzzy extractor. Finally, we introduce the system model and formal scheme definition, and discuss the design goals and the security model.

A. Computational Hardness Assumption

ECC: The nonsingular elliptic curve is defined over the finite field F_q, denoted as $E(F_q)$, where q is a large prime number. $E(F_q)$ comprises points that satisfy the equation $y^2 = x^3 + ax + b \bmod q$, where $a, b \in F_q$ and $4a^3 + 27b^2 \neq 0 \bmod q$. The points on the elliptic curve, including the infinite point O, form an additive group denoted as G. Cryptographic difficulties based on elliptic curves are defined as follows.

Elliptic curve discrete logarithm problem (ECDLP): Finding a scalar $x \in Z_q^*$ that satisfies the equation $Q = xP$, where Q belongs to the elliptic curve group G generated by P, poses a formidable computational challenge.

Elliptic curve Diffie–Hellman problem (ECDHP): Computing abP using the given random points $Q = aP$ and $R = bP$ in the elliptic curve group G with generator P, where $a, b \in Z_q^*$, poses a formidable computational challenge.

B. Chameleon Hash

The CH function [36], introduced by Krawczyk and Rabin, incorporates a trapdoor mechanism that enables efficient collision finding for individuals possessing the trapdoor, while remaining computationally infeasible for those without it. The CH function comprises five key algorithms: Setup, KeyGen, Hash, Verify, and Forge.

1) *Setup(λ):* The algorithm takes the security parameter λ as input and produces the public parameters pp as its output.
2) *KeyGen(pp):* The algorithm takes the public parameters pp as input and generates the public/private key pair $(\mathrm{GK}, \mathrm{VK})$, where GK represents the public key and VK represents the private key.
3) *Hash(GK, m, r):* The algorithm takes the public key GK, message m, and a random value r as inputs and generates the CH.
4) *Verify(GK, m, r, CH):* The algorithm takes the public key GK, message m, the random value r, and CH as inputs and determines the validity of CH.
5) *Forge(VK, m, r, m'):* The algorithm takes the private key VK, the message m, the random value r, and a new message m' as inputs, it computes a new random value r' such that *Verify*$(\mathrm{GK}, m', r', \mathrm{CH}) =$ *Verify*$(\mathrm{GK}, m, r, \mathrm{CH})$.
6) *Collision resistance:* There is no probability polynomial time (PPT) algorithm that on input the public key GK can outputs two pairs (m_1, r_1) and (m_2, r_2) such that $m_1 \neq m_2$ and *Verify*$(\mathrm{GK}, m_1, r_1, \mathrm{CH}) =$ *Verify*$(\mathrm{GK}, m_2, r_2, \mathrm{CH})$.

C. Fuzzy Extractor

The fuzzy extractor [37] consists of two algorithms $(\mathrm{Gen}, \mathrm{Rep})$. Given the biometric BIO, the Gen algorithm generates a biometric key α and the public parameter β. The Rep algorithm can recover the biometric key α if provided with a biometric within the tolerance threshold and the corresponding the public parameter β.

D. System Model of BCCA

The system model of BCCA scheme is depicted in Fig. 1, and it comprises four distinct entities: blockchain, PKG, users, task publisher.

Blockchain: We have adopted a dual-chain architecture comprising a historical chain and an evidence chain. The historical chain is utilized for storing verified signature information to ensure the traceability of history. On the other hand, the evidence chain is dedicated to storing evidence of malicious users. This

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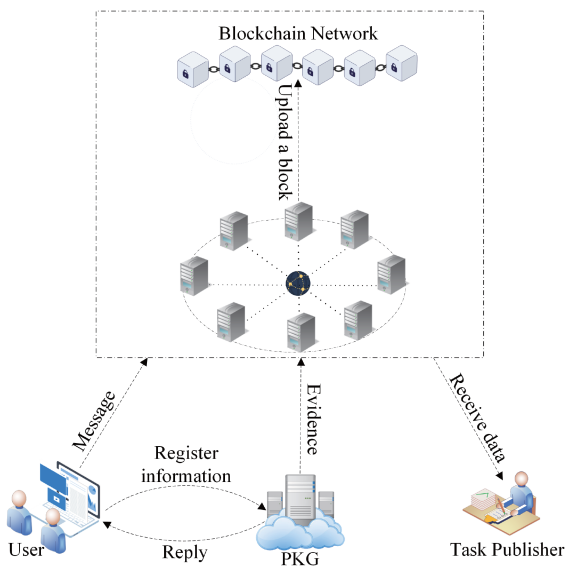

Fig. 1. System model.

dual-chain architecture enables us to enhance information credibility and meet the requirements for recording and addressing malicious activities.

PKG: The PKG is responsible for generating the system's public parameters as well as the corresponding part of the user's private key. It employs a fuzzy extractor to create login authentication credentials for users. Furthermore, as an authoritative entity, the PKG possesses the capability to trace the real identity of users engaged in illegal activities. Additionally, it has the capacity to dynamically modify information stored within the evidence blockchain, enhancing the integrity and reliability of the system.

Task publisher: The task publisher queries and decrypts the ciphertext on the latest block to get the data.

Users: Each user is equipped with an industrial controller, enabling them to interact with sensors and collect data. Subsequently, users generate signatures for the collected data. Moreover, a certificateless system requires users to generate another partial private key themselves.

E. Formal Definition of BCCA

The following presents the formal definition of BCCA.

Definition 1: The BCCA consists of the following seven polynomial-time algorithms: setup, set-private-key, set-secret-value, extract-partial-private-key, login, sign, verify.

Setup: The PKG executes the PPT algorithm. With the security parameter κ as inputs, PKG generates its master secret key s, and system public paramters $\text{params} = \{\mathbb{E}, \mathbb{G}, q, P, P_{\text{pub}}, \text{tpk}, H_{i=1,2,3}\}$, where tpk is the public key of task publisher.

Set-secret-value: The user executes the PPT algorithm. With Params as inputs, the user generates the true identity RID_i, secret key x_i, the corresponding public key upk_i, and the biometric BIO_i. Then, the user sends $\{\text{RID}_i, \text{upk}_i, \text{BIO}_i\}$ to PKG via a secure channel.

Extract-partial-private-key: The PKG executes the PPT algorithm. With $\{\text{params}, \text{RID}_i, \text{BIO}_i, \text{upk}_i\}$ as inputs, the PKG generates a partial private key psk_i, the corresponding public key gpk_i, a random point E_i, the pseudonyms identity ID_i, and login authentication credentials $\{\beta_i, A_i, B_i\}$. Finally, the PKG sends $\{\text{gpk}_i, \text{psk}_i, E_i, \text{ID}_i, \beta_i, A_i, B_i\}$ to the user via a secure channel.

Set-public/private-key: The user executes the PPT algorithm. With $\{\text{params}, \text{psk}_i, \text{gpk}_i\}$ as inputs, the user first verify the validity of the psk_i. Then, the user set $\text{sk}_i = (x_i, \text{psk}_i)$ as his private key, $\text{pk}_i = (\text{upk}_i, \text{gpk}_i)$ as his public key. Finally, the user generates the precomputed secret key set SID_i and the precomputed public key set KID_i.

Login: The industrial controller executes the PPT algorithm. With $\{\text{params}, \text{RID}_i, \text{BIO}_i, \text{PW}_i\}$ as inputs, industrial controller can check whether the login message is legitimate.

Sign: The user executes the PPT algorithm. With $\{\text{params}, m_i\}$ as inputs, the user produces a signature σ_i.

Verify: The consensus server VS_i in the blockchain executes the PPT algorithm. With a group of messages from all users as inputs, the consensus servers perform authentication and integrity checks on these messages. Once a consensus is reached among all the servers, these messages will be uploaded to the consortium blockchain.

F. Design Goals and Security Model

The BCCA scheme faces two types of security threats. First, IIoT communication channels are not secure, and adversaries can monitor the communication channels to read, intercept, replay, and modify critical data transmitted over the channels, as well as generate forged data to successfully pass authentication and integrity checks. Second, adversaries can infer a user's true identity from sensitive data. However, the inability to trace the source of anonymous attacks while protecting real identities could be catastrophic for the system.

Therefore, to effectively implement BCCA in IIoT and address the aforementioned security threats, it is necessary to achieve the following design objectives.

1) *Conditional identity privacy:* The communication should preserve the anonymity of the user's real identity, and only the PKG has the capability to disclose the actual identity of a specific user when necessary.
2) *Dynamic modification:* The system should possess the capability to dynamically modify the evidence chain without altering the hash values of the blocks.
3) *Resistance to man-in-middle attack:* To prevent attackers from impersonating legitimate users, entities are mutually authenticated, ensuring that attackers cannot masquerade as legitimate entities to send messages.
4) *Resistance to replay attack:* Prevent malicious attackers from intercepting messages and repeatedly sending them to the consensus server for illegal purposes.
5) *Resistance to temporary secret leakage attack:* Even if temporary random numbers are leaked, attackers cannot acquire the necessary random factors for authentication.
6) *Resistance to key recover attack:* It is crucial to prevent attackers from deducing users' private keys by collecting batch signatures.
7) *Resistance to forgery attack:* Prevent malicious attackers from forging legitimate signatures to bypass the signature verification phase.
8) *Enhanced performance:* Enhanced performance is an essential prerequisite for employing BCCA in IIoT. In particular, given the limited processing power of the consensus

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server, the computational cost and communication overhead should be as low as possible to satisfy the real-time demands of the IIoT.

As the guarantee of nonforgeability on signatures is crucial for ensuring the correct authentication of data, it stands as the primary security characteristic of the BCCA. Therefore, it is imperative to provide a description of the security model of nonforgeability on signatures in advance. In the initial security model introduced by Al-Riyami et al. [8], there are two types of adversary $\mathcal{A}_1$ and $\mathcal{A}_2$ defined as follows.

1) *Type-I adversary* $\mathcal{A}_1$: An external attacker is unable to obtain the user's partial private key and the system master key, but they have the capability to substitute the user's public key with their own chosen value.
2) *Type-II adversary* $\mathcal{A}_2$: A malicious-but-passive PKG possesses the system master key and has the ability to acquire the user's partial private key, but it lacks the capability to alter the public key of the target user.

In a security model introduced by Huang et al. [38], the *super* adversary is the most powerful and dangerous. For the *super* adversary, who is able to obtains a valid signature, the public key can be replaced with a random number, even if he does not have the corresponding secret key.

A *Super* adversary $\mathcal{A}$ can initiate oracle queries to challenger $\mathcal{C}$ in the following manner.

1) *Create user* (ID_i)*:* The oracle generates and stores all parameters related to a specific user identity in the list L_u. When provided with the identity ID_i as input, $\mathcal{C}$ executes the corresponding algorithms and outputs $(\mathrm{psk}_i, x_i, \mathrm{PK}_i)$. After that, $\mathcal{C}$ sends PK_i to $\mathcal{A}$ and stores $(\mathrm{ID}_i, \mathrm{psk}_i, x_i, \mathrm{PK}_i)$ into the list L_u as an entry.
2) *Extract partial private key* (ID_i)*:* The oracle provides $\mathcal{A}$ with the user's partial private key psk_i in response to its queries.
3) *Extract secret value* (ID_i)*:* The oracle provides $\mathcal{A}$ with the user's secret key x_i in response to its queries.
4) *Replace public key* $(\mathrm{ID}_i, x_i', \mathrm{PK}_i')$*:* The oracle handles $\mathcal{A}$s requests to modify a specific user's public key. $\mathcal{C}$ updates the corresponding entry $(\mathrm{ID}_i, \mathrm{psk}_i, x_i, \mathrm{PK}_i)$ to $(\mathrm{ID}_i, \mathrm{psk}_i, x_i', \mathrm{PK}_i')$. It should be noted that x_i' can be set to $\perp$.
5) *Super sign* (ID_i, m_i)*:* This oracle responds to $\mathcal{A}$s queries for a valid signature σ_i. Upon receiving a message m_i and an identity ID_i, challenger $\mathcal{C}$ generates a valid signature σ_i using the information stored in corresponding list, ensuring that *Verify* $(\mathrm{ID}_i, m_i, \sigma_i, \mathrm{Params}, \mathrm{PK}_i) = 1$.

In this article, we focus on the two types of adversary models. The security model via games between the super adversaries $\mathcal{A}_1$ (or $\mathcal{A}_2$) and $\mathcal{C}_1$ (or $\mathcal{C}_2$) is as follows.

Definition 2: A CLS scheme achieves existential unforgeability under adaptive chosen message attacks (EUF-CMA) when the advantage $Adv_{\mathcal{A}_i}(\lambda)$ of any polynomial-time super adversaries $\mathcal{A}_1$ and $\mathcal{A}_2$ is negligible in both games, where $i = 1, 2$.

Game I. The game is executed between a super Type-I adversary $\mathcal{A}_1$ and a challenger $\mathcal{C}_1$. The game is structured into three distinct phases, described as follows.

1) *Setup*: $\mathcal{C}_1$ executes the *Setup* algorithm, generating public parameters Params and the system master key a. Afterward, $\mathcal{C}_1$ securely retains a while providing Params to $\mathcal{A}_1$.
2) *Queries*: $\mathcal{A}_1$ adaptively submits queries to *create user* oracle, *extract partial private key* oracle, *extract secret value* oracle, *replace public key* oracle, and *super sign* oracle within polynomial time.
3) *Forgery*: After executing polynomial queries during the *Queries* phase, $\mathcal{A}_1$ generates a forged signature pair (m_i^*, σ_i^*) associated with the specific identity ID_i^*.

We posit that $\mathcal{A}_1$ can win in Game I if and only if the following conditions are fulfilled.

1) *Verify* $(\mathrm{ID}_i^*, m_i^*, \sigma_i^*, \mathrm{Params}, \mathrm{PK}_i^*)$=1.
2) The *Super sign* oracle has not received the query (ID_i^*, m_i^*).
3) *Extract partial private key* oracle has not received the query for ID_i^*.

Game II. This game is executed between a super Type-II adversary $\mathcal{A}_2$ and a challenger $\mathcal{C}_2$. The game is structured into three distinct phases, described as follows.

1) *Setup*: $\mathcal{C}_2$ executes the *Setup* algorithm and subsequently provides $\mathcal{A}_2$ with the public parameters Params and the master key a.
2) *Queries*: $\mathcal{A}_2$ adaptively submits queries to *create user* oracle, *extract partial private key* oracle, *extract secret value* oracle, *replace public key* oracle, and *super sign* oracle within polynomial time.
3) *Forgery*: After executing polynomial queries during the *Queries* phase, $\mathcal{A}_2$ generates a forged signature pair (m_i^*, σ_i^*) associated with the specific identity ID_i^*.

We posit that $\mathcal{A}_2$ can win in Game II if and only if the following conditions are fulfilled.

1) *Verify* $(\mathrm{ID}_i^*, m_i^*, \sigma_i^*, \mathrm{Params}, \mathrm{PK}_i^*)= 1$.
2) *Super sign* oracle has not received the query (ID_i^*, m_i^*).
3) *Extract secret value* oracle has not received the query ID_i^*.
4) *Replace public key* oracle has not received the query ID_i^*.

Overall, within our security model, we have considered two types of adversaries. While they exhibit distinct characteristics, they share a common capability, which is the ability to substitute public keys with selected values. The ultimate objective of this substitution behavior is to facilitate the forgery of signatures.

III. BCCA Scheme

In this section, we will introduce our BCCA scheme, which supports user revocation and on-chain modification.

A. Construction of BCCA

This scheme consists of the following seven polynomial-time algorithms: setup, set-private-key, extract-partial-private-value, set-key, login, sign, verify. The corresponding cryptographic notation can be found in the Nomenclature.

Setup: In this phase, PKG generates the system's public parameters based on a security parameter κ.

1) Given a security parameter κ, PKG selects an elliptic curve $\mathbb{E}$ over a finite field Z_p, and chooses an additive group $\mathbb{G}$ with order q derived from the elliptic curve $\mathbb{E}$.
2) PKG generates a generator P of $\mathbb{G}$ and randomly chooses $s \in Z_q^*$ as the master key. It then computes the system pubic key $P_{\text{pub}} = s \cdot P$.
3) PKG randomly chooses the private key $y \in Z_q^*$ for a task publisher and computes the corresponding public key $\mathrm{tpk} = y \cdot P$.
4) PKG determines three collision resistant hash functions: $H_i : \{0,1\}^* \rightarrow Z_q^*$, where $i = 1, 2, 3$.

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5) Finally, PKG publishes the system public parameters $\text{params} = \{\mathbb{E}, \mathbb{G}, q, P, P_{\text{pub}}, \text{tpk}, H_{i=1,2,3}\}$.

Set-secret-value: In this phase, U_i registers with the PKG via a secure channel.

1) U_i chooses a random number $x_i \in Z_q^*$ as his/her secret key, and computes the $\text{upk}_i = x_i \cdot P$.
2) U_i take as inputs his/her real identity RID_i and password PW_i, and computes $\text{UPW}_i = H_1(\text{RID}_i, \text{PW}_i)$.
3) U_i sends the registration information $\text{Reg}_i = (\text{upk}_i, \text{RID}_i, \text{UPW}_i, \text{BIO}_i)$ to PKG via a secure channel, where BIO_i denotes the biometric.

Extract-partial-private-key: In this phase, PKG generates the U_is partial private key psk_i, and generates the anonymous identity ID_i.

1) PKG chooses a random number $d_i \in Z_q^*$, and computes $E_i = d_i \cdot P$ and $E_i^* = s \cdot E_i$. Then, PKG computes the pseudonyms identity $\text{ID}_i = \text{Enc}_{E_{i,x}^*}(\text{RID}_i)$, where $E_{i,x}^*$ denotes the x-coordinate of the point E_i^*.
2) PKG chooses a random number $k_i \in Z_q^*$, and computes $\text{gpk}_i = k_i \cdot P$. Then, the PKG computes the U_is partial private key $\text{psk}_i = k_i + s \cdot h_{1,i}$, where $h_{1,i} = H_1(\text{ID}_i, \text{gpk}_i, \text{upk}_i, P_{\text{pub}}, E_i)$.
3) PKG computes $\text{Gen}(\text{BIO}_i) = (\alpha_i, \beta_i)$. Then, it computes $A_i = \text{psk}_i \cdot \text{UPW}_i$, and $B_i = H_1(\alpha_i, \text{UPW}_i, \text{psk}_i)$.
4) PKG sends the reply $\{\text{gpk}_i, \text{psk}_i, E_i, \text{ID}_i, \beta_i, A_i, B_i\}$ to the U_i via a secure channel.

Set-public/private-key: In this phase, U_i sets his/her private key set and public key. Additionally, U_i generates a presecret set in advance.

1) U_i verifies $\text{psk}_i \cdot P = \text{gpk}_i + h_{1,i} \cdot P_{\text{pub}}$. If it holds, U_i sets $\text{pk}_i = \{\text{upk}_i, \text{gpk}_i\}$ as his public key, $\text{sk}_i = (x_i, \text{psk}_i)$ as his private key.
2) U_i chooses n random numbers $v_i = \{v_{i,j}\}_{j \in [n]}$, where $v_{i,j} \in Z_q^*$. Then, U_i computes the precomputed secret key set $\text{SID}_i = \{\text{SID}_{i,j}\}_{j \in [n]}$, where $\text{SID}_{i,j} = v_{i,j} x_i + H_1(\text{RID}_i, \beta_i)$. Finally, the corresponding precomputed public key set is $\text{KID}_i = \{\text{KID}_{i,j}\}_{j \in [n]}$, where $\text{KID}_{i,j} = \text{SID}_{i,j} \cdot P$.
3) U_i securely stores $\{A_i, B_i, \beta_i, \text{SID}_i, \text{KID}_i\}$ within the industrial controller.

Remark 1: We generates a pregenerated temporary secret set, which can be directly retrieved from the local database during signature generation. This approach avoids the need for generating a new temporary secret value for each signature, reducing the time and computational cost required for the signing process.

Login: In this phase, prior to accessing the industrial controller, user authentication is required.

1) U_i inserts the smart card and provides BIO_i, PW_i, and RID_i inputs to the industrial controller.
2) Industrial controller computes $\alpha_i = \text{Rep}(\text{BIO}_i, \beta_i)$, and $\text{UPW}_i = H_1(\text{PW}_i, \text{RID}_i)$. Then, it computes $\text{psk}_i = A_i \cdot \text{UPW}_i^{-1}$.
3) Industrial controller checks $B_i = H_1(\alpha_i, \text{UPW}_i, \text{psk}_i)$ whether holds. If it holds, U_i can log in the industrial controller; otherwise, the login is denied.

Remark 2: The integration of fuzzy extractors and hash functions provides an effective defense against offline password guessing attacks.

Sign: In this phase, U_i controls the sensor to collect data and then encrypts his/her data m_i. Subsequently, U_i generates the corresponding signature σ_i.

1) U_i chooses a random number $q_i \in Z_q^*$, computes $Q_i = q_i \cdot P$ and encrypts m_i as $c_i = m_i \oplus H_1(q_i \cdot \text{tpk})$.
2) U_i computes $h_{2,i} = H_2(\text{ID}_i, \text{pk}_i, \text{KID}_{i,k}, Q_i, P_{\text{pub}})$, and $h_{3,i} = H_3(\text{ID}_i, c_i, \text{pk}_i, P_{\text{pub}}, T_i)$, where T_i is the current timestamp. Finally, U_i computes signature $\sigma_i = \text{psk}_i + h_{2,i}\text{SID}_{i,k} + h_{3,i} x_i$.
3) Finally, U_i sends $\text{Msg}_i = \{\sigma_i, \text{ID}_i, E_i, \text{pk}_i, \text{KID}_{i,k}, c_i, Q_i, T_i\}$, to the consensus server VS_i in the consortium blockchain.

Remark 3: $\text{SID}_{i,k}$ represents the kth element of set SID_i, similar to $\text{KID}_{i,k}$ mentioned in the text.

Verify: In this phase, all VS_i check the correctness of the message Msg_i and reach a consensus.

1) VS_i obtains the current time T_{cur} and checks the freshness of T_i. If $|T_i - T_{\text{cur}}| \leq \Delta T$, continues; otherwise, discards.
2) VS_i computes $h_{1,i} = H_1(\text{ID}_i, \text{pk}_i, P_{\text{pub}}, E_i)$, $h_{2,i} = H_2(\text{ID}_i, \text{pk}_i, \text{KID}_{i,j}, Q_i, P_{\text{pub}})$, and $h_{3,i} = H_3(\text{ID}_i, c_i, \text{pk}_i, P_{\text{pub}}, T_i)$. Then, it verifies $\sigma_i \cdot P = \text{gpk}_i + h_{1,i} P_{\text{pub}} + h_{2,i}\text{KID}_{i,k} + h_{3,i}\text{upk}_i$ whether holds. If it holds, accept the message $\text{Msg}_i = (\sigma_i, \text{ID}_i, E_i, \text{pk}_i, \text{KID}_{i,k}, c_i, Q_i, T_i)$; otherwise, reject it.
3) Finally, the task publisher queries Msg_i on the latest block and decrypts the c_i as $m_i = c_i \oplus H_1(yQ_i)$.

Remark 4: Storing the Msg_i on the blockchain also provides convenience for future auditing and supervision, which can better maintain security and trust.

B. Supporting Revocation

When a historical audit on the blockchain reveals the presence of malicious users, auditors can report specific ID_i to the PKG along with corresponding evidence Evid_i. The PKG can decrypt the user's real identity and record the relevant evidence on the evidence chain.

1) The PKG computes $E_i^* = s \cdot E_i$, and decrypt the ID_i to obtain RID_i.
2) The PKG chooses a random number $\theta_i \in Z_q^*$, computes $\text{ck}_i = H_1(s, \theta_i)$ as a trapdoor, and computes $\text{HK}_i = \text{ck}_i \cdot P$.
3) the PKG chooses a random number $\zeta_i \in Z_q^*$ and another random number $j_i \in Z_q^*$ as a salt. Finally, the PKG computes $\text{CH}_i = (\zeta_i \cdot \text{ck}_i + H_1(j_i, \text{cred}_i, \text{HK}_i)) \cdot P$, where $\text{cred}_i = \{\text{ID}_i, \text{RID}_i, \text{Evid}_i\}$. Herein, CH_i is the CH value for U_is cred_i.
4) By considering Cred_i as a transaction and CH_i as the hash of each transaction, a Merkle tree can be constructed. This enables PKG to create new block. Finally, PKG uploads the new block to the evidence chain.
5) PKG stores the tuple $\{\text{HK}_i, j_i, \theta_i, \zeta_i, \text{ID}_i, \text{RID}_i\}$ into its database D_{evd}.

Remark 5: By disclosing the identity information of malicious users, other participants can identify and avoid interacting with those users, thereby preventing fraudulent activities and enhancing the security of the system. In addition, in contrast to the traditional elliptic curve-based CH algorithm as proposed in [39] and [40], our approach utilizes only one private key, which not only saves storage cost but also enhances efficiency. Additionally, it incorporates salting to improve collision resistance.

C. Supporting On-Chain Modification

1) PKG searches for the original random numbers ζ_i and θ_i, as well as the original salt j_i in D_{evd}, and computes the trapdoor $\text{ck}_i = H_1(s, \theta_i)$.

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2) PKG generates a new $\text{cred}'_i = \{\text{ID}_i, \text{RID}_i, \text{Evid}'_i\}$. Then, PKG can compute $\zeta'_i = \text{ck}_i^{-1}(H_1(j_i, \text{cred}_i, \text{HK}_i) - H_1(j_i, \text{cred}'_i, \text{HK}_i)) + \zeta_i$. Therefore, the modification of cred_i can be performed without impacting the hash value of the block.

Remark 6: The CH technology provides the flexibility to update and correct the evidence chain, enabling the handling of erroneous revocation actions and the supplementation of new evidence.

IV. Security Analysis

In this section, we provide a comprehensive analysis of the scheme's correctness and security. We approach this analysis from three different perspectives: correctness, formal security analysis, and informal security analysis.

A. Correctness

First, the verification for a single signature is elaborated as follows:

$$\begin{aligned}\sigma_i \cdot P &= (\text{psk}_i + h_{2,i}\text{SID}_{i,k} + h_{3,i}x_i) \cdot P \\ &= (k_i + h_{1,i}s) \cdot P + h_{2,i}\text{SID}_{i,k} \cdot P + h_{3,i}x_i \cdot P \\ &= \text{gpk}_i + h_{1,i}P_{\text{pub}} + h_{2,i}\text{SID}_{i,k} + h_{3,i}\text{upk}_i.\end{aligned}$$

Second, the VS_i verifies batch signature. To ensure nonrepudiation and prevent the confusion attack, the small exponent test [41] has been used in the batch verification phase. VS_i chooses $\lambda = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}$, where λ_i is randomly selected in $[1, 2^t]$, and t is a low-overhead integer parameter with minimal impact on system performance. Thus, the batch authentication of multiple messages is given by

$$\begin{aligned}\left(\sum_{i=1}^{n} \lambda_i \sigma_i\right) \cdot P &= \sum_{i=1}^{n} \lambda_i (\text{psk}_i + h_{2,i}\text{SID}_{i,k} + h_{3,i}x_i) \cdot P \\ &= \sum_{i=1}^{n} \lambda_i \left((k_i + \text{sh}_{1,i}) + h_{2,i}\text{SID}_{i,k}\right. \\ &\quad \left. + h_{3,i}x_i\right) \cdot P \\ &= \sum_{i=1}^{n} \lambda_i (\text{gpk}_i) + \sum_{i=1}^{n} \lambda_i (h_{1,i}P_{\text{pub}}) \\ &\quad + \sum_{i=1}^{n} \lambda_i (h_{2,i}\text{KID}_{i,k}) + \sum_{i=1}^{n} \lambda_i (h_{3,i}\text{upk}_i).\end{aligned}$$

Third, it should be noted that if the PKG intends to update the Cred_i of U_i on the blockchain, this can be accomplished since the following equation satisfy:

$$\text{CH}_i = \zeta_i \cdot \text{HK}_i + H_1(j_i, \text{cred}_i, \text{HK}_i) \cdot P \tag{1}$$

$$\text{CH}_i = \zeta'_i \cdot \text{HK}_i + H_1(j_i, \text{cred}'_i, \text{HK}_i) \cdot P. \tag{2}$$

Using (1) and (2), the following equation holds:

$$\begin{aligned}&\zeta_i \cdot \text{ck}_i \cdot P + H_1(j_i, \text{cred}_i, \text{HK}_i) \cdot P \\ &= \zeta'_i \cdot \text{ck}_i \cdot P + H_1(j_i, \text{cred}'_i, \text{HK}_i) \cdot P.\end{aligned}$$

Then, PKG can compute $\zeta'_i = \text{ck}_i^{-1}(H_1(j_i, \text{cred}_i, \text{HK}_i) - H_1(j_i, \text{cred}'_i, \text{HK}_i)) + \zeta_i$.

B. Formal Security Analysis

In this section, we prove the CLS scheme can achieve EUF-CMA secure against Type-I adversary $\mathcal{A}_1$ and Type-II adversary $\mathcal{A}_2$.

Theorem 1: In the random oracle, the proposed scheme can effectively withstand attacks from Type-I adversary $\mathcal{A}_1$ under the assumption that ECDLP problem is hard.

Proof: We assume the existence of a Type-I adversary $\mathcal{A}_1$ that can successfully win Game I with a non-negligible probability ϵ. Based on this assumption, we construct a challenger $\mathcal{C}_1$ that can solve the ECDLP problem with a non-negligible probability. With the cooperation of adversary $\mathcal{A}_1$, challenger $\mathcal{C}_1$ attempts to solve the ECDLP problem by finding the value of a in the given instance $(G, P, Q = aP)$.

Setup: $\mathcal{C}_1$ initiates the interaction by setting $P_{\text{pub}} = aP$ and providing the system's public parameters to $\mathcal{A}_1$. Additionally, $\mathcal{C}_1$ randomly chooses an identity ID_t as the target. Subsequently, $\mathcal{A}_1$ proceeds to perform the following queries.

Hash query: $\mathcal{A}_1$ is permitted to issue polynomial queries to $\mathcal{C}_1$. To facilitate simulation, $\mathcal{C}_1$ maintains various lists during the query phase. These include a user list L_u and three hash lists L_{H_1}, L_{H_2}, and L_{H_3}. These three hash lists serve as storage for query values associated with $h_{1,i}$, $h_{2,i}$, and $h_{3,i}$.

$h_{1,i}$ queries: $\mathcal{A}_1$ provides the inputs $(\text{ID}_i, \text{pk}_i, P_{\text{pub}}, E_i)$ to $\mathcal{C}_1$. If $(\text{ID}_i, \text{pk}_i, P_{\text{pub}}, E_i, h_{1,i})$ is found in the list L_{H_1}, $\mathcal{C}_1$ returns $h_{1,i}$ to $\mathcal{A}_1$. Otherwise, $\mathcal{C}_1$ generates a random number $h_{1,i} \in Z_q^*$, adds $(\text{ID}_i, \text{pk}_i, P_{\text{pub}}, E_i, h_{1,i})$ to L_{H_1}, and returns $h_{1,i}$ to $\mathcal{A}_1$.

$h_{2,i}$ queries: $\mathcal{A}_1$ provides the inputs $(\text{ID}_i, \text{pk}_i, \text{KID}_{i,k}, Q_i, P_{\text{pub}})$ to $\mathcal{C}_1$. If $(\text{ID}_i, \text{pk}_i, \text{KID}_{i,k}, Q_i, P_{pub}, h_{2,i})$ is found in the list L_{H_2}, $\mathcal{C}_1$ returns $h_{2,i}$ to $\mathcal{A}_1$. Otherwise, $\mathcal{C}_1$ generates a random number $h_{2,i} \in Z_q^*$, adds $(\text{ID}_i, \text{pk}_i, \text{KID}_{i,k}, Q_i, P_{\text{pub}}, h_{2,i})$ to L_{H_2}, and returns $h_{2,i}$ to $\mathcal{A}_1$.

$h_{3,i}$ queries: $\mathcal{A}_1$ provides the inputs $(\text{ID}_i, c_i, \text{pk}_i, P_{\text{pub}}, T_i)$ to $\mathcal{C}_1$. If $(\text{ID}_i, c_i, \text{pk}_i, P_{\text{pub}}, T_i, h_{3,i})$ is found in the list L_{H_3}, $\mathcal{C}_1$ returns $h_{3,i}$ to $\mathcal{A}_1$. Otherwise, $\mathcal{C}_1$ generates a random number $h_{3,i} \in Z_q^*$, $\mathcal{C}_1$ adds $(\text{ID}_i, c_i, \text{pk}_i, P_{\text{pub}}, T_i, h_{3,i})$ to L_{H_3}, and returns $h_{3,i}$ to $\mathcal{A}_1$.

Create user oracle: When receiving this query with ID_i from $\mathcal{A}_1$, $\mathcal{C}_1$ verifies the presence of the corresponding entry in the list L_{H_1}, and returns $\text{pk}_i = (\text{gpk}_i, \text{upk}_i)$ to $\mathcal{A}_1$ if the entry exists. However, if the requested entry is not found, $\mathcal{C}_1$ proceeds with the following decision procedure.

1) If $\text{ID}_i \neq \text{ID}_t$, $\mathcal{C}_1$ randomly selects $\text{psk}_i, x_i, h_{1,i} \in Z_q^*$, computes $\text{gpk}_i = \text{psk}_i P - h_{1,i}P_{\text{pub}} \bmod q$, $\text{upk}_i = x_i P$.
2) If $\text{ID}_i = \text{ID}_t$, $\mathcal{C}_1$ selects $k_i, x_i, h_{1,i} \in Z_q^*$ and lets $\text{gpk}_i = k_i P$, $\text{upk}_i = x_i P$, and $\text{psk}_i = \perp$.

At last, $\mathcal{C}_1$ outputs $\text{pk}_i = (\text{gpk}_i, \text{upk}_i)$ to $\mathcal{A}_1$ and adds $(\text{ID}_i, \text{gpk}_i, \text{upk}_i, psk_i, x_i, h_{1,i})$ to the list L_u.

Extract partial private key oracle: When receiving the query with ID_i from $\mathcal{A}_1$, $\mathcal{C}1$ takes the following actions.

1) If $\text{ID}_i = \text{ID}_t$, $\mathcal{C}_1$ aborts the game.
2) If $\text{ID}_i \neq \text{ID}_t$, $\mathcal{C}_1$ searches the list L_u and returns psk_i to $\mathcal{A}_1$ if found.
3) If there is no corresponding entry in L_u, $\mathcal{C}_1$ queries the create user oracle with ID_i, and returns psk_i to $\mathcal{A}_1$.

Extract secret value oracle: When receiving this query ID_i from $\mathcal{A}_1$, $\mathcal{C}_1$ performs the following actions.

1) if $\text{ID}_i = \text{ID}_t$, $\mathcal{C}_1$ aborts the game.
2) If $\text{ID}_i \neq \text{ID}_t$, $\mathcal{C}_1$ searches the list L_u and returns x_i to $\mathcal{A}_1$ if found.

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3) If there is no corresponding entry in L_u, $\mathcal{C}_1$ queries create user oracle with ID_i, and returns x_i to $\mathcal{A}_1$.

Replace public key oracle: When receiving this query with $(\mathrm{ID}_i, x_i', \mathrm{pk}_i')$ from $\mathcal{A}_1$, where $\mathrm{pk}_i' = (\mathrm{gpk}_i', \mathrm{upk}_i')$, $\mathcal{C}_1$ looks into L_u to get the entry $(\mathrm{ID}_i, \mathrm{gpk}_i, \mathrm{upk}_i, p\mathrm{sk}_i, x_i, h_{1,i})$ and replaces $\mathrm{pk}_i = (\mathrm{gpk}_i, \mathrm{upk}_i)$ with pk_i'.

Super sign oracle: When Receiving this query with (ID_i, c_i) from $\mathcal{A}_1$, $\mathcal{C}_1$ first determines whether $\mathrm{ID}_i = \mathrm{ID}_t$ holds.

1) If $\mathrm{ID}_i \neq \mathrm{ID}_t$ and $x_i \neq \perp$ (indicating that the public key has not been replaced), $\mathcal{C}_1$ randomly selects $\mathrm{SID}_{i,k} \in Z_q^*$, traverses lists $L_u, L_{H_1}, L_{H_2}, L_{H_3}$ to get $\mathrm{psk}_i, x_i, h_{2,i}, h_{3,i}$ and computes $\sigma_i = p\mathrm{sk}_i + h_{2,i}\mathrm{SID}_{i,k} + h_{3,i}x_i \bmod q$, and returns to $\mathcal{A}_1$.
2) If $\mathrm{ID}_i = \mathrm{ID}_t$ or $x_i = \perp$, $\mathcal{C}_1$ randomly chooses $\sigma_i, h_{1,i}, h_{2,i}, h_{3,i} \in Z_q^*$, computes the $\mathrm{KID}_{i,k} = h_{2,i}^{-1}(\sigma_i P - (\mathrm{gpk}_i + h_{1,i}P_{\mathrm{pub}} + h_{3,i}\mathrm{upk}_i))$, and returns to $\mathcal{A}_1$.

Forgery: $\mathcal{A}_1$ outputs forged signatures. The ID_i^* is not submitted to extract partial private key oracle and (c_i^*, ID_i^*) has never been queries to super sign oracle. If $\mathrm{ID}_i^* \neq \mathrm{ID}_t$, $\mathcal{C}_1$ aborts the game. Otherwise, $\mathcal{C}_1$ refers to the list L_u, L_{H_1}, L_{H_2}, and L_{H_3}. Due to the forking lemma [42], $\mathcal{A}_1$ succeeds to forge the two legitimate signatures $(\mathrm{ID}_i^*, c_i^*, \sigma_i^*, \mathrm{KID}_{i,k}^*)$ and $(\mathrm{ID}_i^*, c_i^*, \sigma_i', \mathrm{KID}_{i,k}^*)$ in polynomial time under distinct $h_{1,i}$, which satisfy the following:

$$\sigma_i^* P = \mathrm{gpk}_i^* + h_{1,i}^{(1)} P_{\mathrm{pub}} + h_{2,i}^* \mathrm{KID}_{i,k}^* + h_{3,i}^* \mathrm{upk}_i^* \tag{3}$$

$$\sigma_i' P = \mathrm{gpk}_i^* + h_{1,i}^{(2)} P_{\mathrm{pub}} + h_{2,i}^* \mathrm{KID}_{i,k}^* + h_{3,i}^* \mathrm{upk}_i^*. \tag{4}$$

Using (3) minus (4), we drive the following derivation:

$$\begin{aligned}\sigma_i^* P - \sigma_i' P &= \mathrm{gpk}_i^* + h_{1,i}^{(1)} P_{\mathrm{pub}} + h_{2,i}^* \mathrm{KID}_{i,k}^* + h_{3,i}^* \mathrm{upk}_i^* \\ &\quad - (\mathrm{gpk}_i^* + h_{1,i}^{(2)} P_{\mathrm{pub}} + h_{2,i}^* \mathrm{KID}_{i,k}^* + h_{3,i}^* \mathrm{upk}_i^*) \\ &= (h_{1,i}^{(1)} - h_{1,i}^{(2)}) P_{\mathrm{pub}} \\ &= (h_{1,i}^{(1)} - h_{1,i}^{(2)}) aP.\end{aligned}$$

Moreover, based on the given equation, $\mathcal{C}_1$ can derive the value $a = (\sigma_i^* - \sigma_i')(h_{1,i}^{(1)} - h_{1,i}^{(2)})^{-1}$, which solves the ECDLP instance. Next, we analyze the probability of $\mathcal{C}_1$ winning in game I. $\mathcal{C}_1$ succeeds if the following three events occur simultaneously.

1) E_1: When $\mathcal{A}_1$ queries extract partial private key oracle, $\mathcal{C}_1$ continues the game without aborting.
2) E_2: σ_i^* is a valid forgery on (c_i^*, ID_i^*).
3) E_3: Regarding the forged signature (c_i^*, ID_i^*) submitted by $\mathcal{A}_1$ during the forgery phase, it holds that $\mathrm{ID}_i^* = \mathrm{ID}_t$.

From the analysis process of the game, we could have $\Pr[E_2|E_1] \geqslant \epsilon$, $\Pr[E_1] \geq (1 - \frac{1}{q_{\mathrm{cu}}})^{q_{eppk}}$, and $\Pr[E_3|E_1E_2] \geq \frac{1}{q_{\mathrm{cu}}}$, where q_{cu} and q_{eppk} represent the maximum number of create user oracle queries and extract partial private key oracle queries. The probability of $\mathcal{C}_1$ successfully solve the ECDLP problem is as follows:

$$\begin{aligned}\epsilon_1 &= \Pr[E_1E_2E_3] \\ &= \Pr[E_1]Pr[E_2|E_1]\Pr[E_3|E_1E_2] \\ &\geq \left(1 - \frac{1}{q_{\mathrm{cu}}}\right)^{q_{eppk}} \frac{1}{q_{\mathrm{cu}}}\epsilon.\end{aligned}$$

Theorem 2: In the random oracle, the proposed scheme can effectively withstand attacks from Type-II adversary $\mathcal{A}_2$ under the assumption that ECDLP problem is hard.

Proof: We assume the existence of a *Type-II adversary* $\mathcal{A}_2$ that can successfully win Game I with a non-negligible probability ϵ. Based on this assumption, we construct a challenger $\mathcal{C}_2$ that can solve the ECDLP problem with a non-negligible probability. With the cooperation of adversary $\mathcal{A}_2$, challenger $\mathcal{C}_2$ attempts to solve the ECDLP problem by finding the value of a in the given instance $(G, P, Q = aP)$.

Setup: $\mathcal{C}_2$ initiates the interaction by setting $\mathrm{upk}_i = a_i P$ and providing the system's public parameters to $\mathcal{A}_2$. Additionally, $\mathcal{C}_2$ randomly chooses an identity ID_t as the target. Subsequently, $\mathcal{A}_2$ proceeds to perform the following queries.

The hash query is the same as the game I except for create user oracle.

Create user oracle: When $\mathcal{C}_2$ receives a query with identity ID_i, it follows the following procedure.

1) $\mathcal{C}_2$ verifies the existence of the corresponding entry in the list L_{H_1}. If the entry exists, $\mathcal{C}2$ returns $\mathrm{pk}_i = (\mathrm{gpk}_i, \mathrm{upk}_i)$ to $\mathcal{A}_2$.
2) If the entry does not exist, $\mathcal{C}_2$ randomly selects $k_i, h_{1,i} \in Z_q^*$, and computes $\mathrm{gpk}_i = k_i P$, $\mathrm{psk}_i = k_i + h_{1,i}s \bmod q$. If $\mathrm{ID}_i \neq \mathrm{ID}_t$, $\mathcal{C}_2$ randomly selects $a_i \in Z_q^*$, computes $\mathrm{upk}_i = a_i P$. If $\mathrm{ID}_i = \mathrm{ID}_t$, $\mathcal{C}_2$ sets $\mathrm{upk}_i = a_i P$ and $a_i = \perp$.

Finally, $\mathcal{C}_2$ returns $\mathrm{pk}_i = (\mathrm{gpk}_i, \mathrm{upk}_i)$ and adds $(\mathrm{ID}_i, \mathrm{gpk}_i, \mathrm{upk}_i, \mathrm{psk}_i, a_i, h_{1,i})$ to the list L_u.

Extract partial private key oracle: When receiving this query with ID_i from $\mathcal{A}_2$, $\mathcal{C}_2$ performs the following actions.

1) If $\mathrm{ID}_i = \mathrm{ID}_t$, $\mathcal{C}_2$ aborts the game.
2) If the corresponding entry exists in the list L_{H_1}, $\mathcal{C}_2$ retrieves a_i, and returns it to $\mathcal{A}_2$.
3) If the entry does not exist, $\mathcal{C}_2$ queries the create user oracle with identity ID_i, and returns a_i to $\mathcal{A}_2$.

Replace public key oracle: When receiving this query with $(\mathrm{ID}_i, \mathrm{pk}_i')$ from $\mathcal{A}_2$, where $\mathrm{pk}_i' = (\mathrm{upk}_i', \mathrm{gpk}_i')$, the following actions take place.

1) If $\mathrm{ID}_i = \mathrm{ID}_t$, $\mathcal{C}_2$ aborts the game.
2) If $\mathrm{ID}_i \neq \mathrm{ID}_t$, $\mathcal{C}_2$ looks into L_u to get the entry $(\mathrm{ID}_i, \mathrm{upk}_i, \mathrm{gpk}_i, \mathrm{psk}_i, a_i, h_{1,i})$ and replaces $\mathrm{pk}_i = (\mathrm{gpk}_i, \mathrm{upk}_i)$ with pk_i'.

Super sign oracle: When Receiving this query with (ID_i, c_i) from $\mathcal{A}_2$, $\mathcal{C}_2$ first determines whether $\mathrm{ID}_i = \mathrm{ID}_t$ holds.

1) If $\mathrm{ID}_i \neq \mathrm{ID}_t$ and $a_i \neq \perp$, $\mathcal{C}_2$ randomly chooses $\mathrm{SID}_{i,k} \in Z_q^*$, traverses lists L_u, L_{H_1}, L_{H_2}, and L_{H_3} to get $\mathrm{psk}_i, a, h_{2,i}, h_{3,i}$, and computes $\sigma_i = psk_i + h_{2,i}\mathrm{SID}_{i,k} + h_{3,i}a_i \bmod q$, and returns to $\mathcal{A}_2$.
2) If $\mathrm{ID}_i = \mathrm{ID}_t$ or $a_i = \perp$, $\mathcal{C}_2$ randomly chooses $\sigma_i, h_{1,i}, h_{2,i}, h_{3,i} \in Z_q^*$, computes the $\mathrm{KID}_{i,k} = h_{2,i}^{-1}(\sigma_i \cdot P - (\mathrm{gpk}_i + h_{1,i}P_{\mathrm{pub}} + h_{3,i}\mathrm{upk}_i))$, and returns to $\mathcal{A}_2$.

Forgery: $\mathcal{A}_2$ outputs forged signatures. The ID_i^* is not submitted to extract partial private key oracle, replace public key oracle, and (c_i^*, ID_i^*) has never been queries to super sign oracle. If $\mathrm{ID}_i^* \neq \mathrm{ID}_t$, $\mathcal{C}_2$ aborts the game. Otherwise, $\mathcal{C}_2$ refers to the list L_1, L_{H_2}, and L_{H_3}. Due to the forking lemma [42], $\mathcal{A}_2$ succeeds to forge the two legitimate signatures $(\mathrm{ID}_i^*, c_i^*, \sigma_i^*, \mathrm{KID}_{i,k}^*)$ and $(\mathrm{ID}_i^*, c_i^*, \sigma_i', \mathrm{KID}_{i,k}^*)$ in polynomial time under distinct $h_{3,i}$, which satisfy

$$\sigma_i^* P = \mathrm{gpk}_i^* + h_{1,i}^* P_{\mathrm{pub}} + h_{2,i}^* \mathrm{KID}_{i,k}^* + h_{3,i}^{(1)} \mathrm{upk}_i^* \tag{5}$$

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$$\sigma_i' P = \text{gpk}_i^* + h_{1,i}^* P_{\text{pub}} + h_{2,i}^* \text{KID}_{i,k}^* + h_{3,i}^{(2)} \text{upk}_i^*. \quad (6)$$

Using (5) minus (6), we drive the following derivation:

$$\begin{aligned}\sigma_i^* P - \sigma_i' P &= \text{gpk}_i^* + h_{1,i}^* P_{\text{pub}} + h_{2,i}^* \text{KID}_{i,k}^* + h_{3,i}^{(1)} \text{upk}_i^* \\ &\quad - (\text{gpk}_i^* + h_{1,i}^* P_{\text{pub}} + h_{2,i}^* \text{KID}_{i,k}^* + h_{3,i}^{(2)} \text{upk}_i^*) \\ &= (h_{3,i}^{(1)} - h_{3,i}^{(2)}) \text{upk}_i \\ &= (h_{3,i}^{(1)} - h_{3,i}^{(2)}) a_i P.\end{aligned}$$

Moreover, based on the given equation, $\mathcal{C}_2$ can derive the value $a_i = (\sigma_i^* - \sigma_i')(h_{3,i}^{(1)} - h_{3,i}^{(2)})^{-1}$, which solves the ECDLP instance. Next, we analyze the probability of $\mathcal{C}_2$ winning in game II. $\mathcal{C}_2$ succeeds if the following three events occur simultaneously.

1) E_1: When $\mathcal{A}_2$ queries extract secret value oracle and replace public key oracle, $\mathcal{C}_2$ continues the game without aborting.
2) E_2: σ_i^* is a valid forgery for (c_i^*, ID_i^*).
3) E_3: For the forged signature (c_i^*, ID_i^*) submitted by $\mathcal{A}_2$ in forgery phase, we have $\text{ID}_i^* = \text{ID}_t^*$.

From the analysis process of the game, we could have $\Pr[E_2|E_1] \geq \epsilon$, $\Pr[E_1] \geq (1 - \frac{1}{q_{\text{cu}}})^{q_{\text{esv}}}(1 - \frac{1}{q_{\text{cu}}})^{q_{rpk}}$, and $\Pr[E_3|E_1E_2] \geq \frac{1}{q_{\text{cu}}}$, where q_{cu}, q_{esv}, and q_{rpk} represent the maximum number of create user oracle queries, extract secret value oracle queries, and replace public key oracle queries. The probability of $\mathcal{C}_2$ successfully solve the ECDLP problem is as follows:

$$\begin{aligned}\epsilon_2 &= \Pr[E_1E_2E_3] \\ &= \Pr[E_1]\Pr[E_2|E_1]Pr[E_3|E_1E_2] \\ &\geq \left(1 - \frac{1}{q_{\text{cu}}}\right)^{q_{\text{esv}}+q_{\text{rpk}}} \frac{1}{q_{\text{cu}}}\epsilon.\end{aligned}$$

C. Informal Security Analysis

1) Conditional Identity Anonymity: The pseudoidentity is constructed as $\text{ID}_i = \text{Enc}_{E_{i,x}^*}(\text{RID}_i)$. Due to the computational infeasible of recovering $E_i^* = \text{sd}_i \cdot P$ from the ECDH hardness assumption, adversary are unable to obtain the real identity RID_i. Specifically, using the master key s, the PKG can compute $E_i^* = sE_i$ and decrypt ID_i to obtain RID_i. Hence, our scheme achieves conditional identity anonymity.

2) Resistance to Man-in-Middle Attack: In this scheme, mutual authentication is achieved between users and consensus servers. An attacker cannot impersonate a legitimate entity or send a valid message, Hence, our scheme is resilient against man-in-the-middle attack.

3) Resistance to Replay Attack: After receiving the Msg_i from U_i, the VS_i first checks the freshness of the Msg_i. T_i indicates the timestamp of Msg_i sent by U_i and T_{cur} indicates the timestamp of Msg_i received by VS_i. If $|T_{\text{cur}} - T_i| \leqslant \Delta T$, then VS_i will receive this message Msg_i, otherwise, VS_i rejects Msg_i, where ΔT denotes the delay that the system can tolerate. Hence, our CLS scheme can resist replay attack.

4) Resistance to Temporary Secret Leakage Attack: In this scheme, whenever a user generates a checksum $\text{SID}_{i,j}$, it is combined with the long-term secrets RID_i, β_i and x_i. Consequently, even the temporary secret leakage, adversaries are unable to reconstruct $\text{SID}_{i,j}$. Hence, our scheme can resist temporary secret leakage attack.

5) Resistance to Key Recover Attack: During the signature generation process, we randomly select $S\text{ID}_{i,j}$ from a precomputed secret key set and subsequently couple it with private keys. This random linkage between private keys and signatures precludes adversaries from inferring users' private keys through signature collection. Hence, our scheme can resist key recover attack.

6) Resistance to Forgery Attack: On the basis of Theorems 1 and 2 mentioned in Section IV-B, we can know that the advantage of adversary $\mathcal{A}_1$ and adversary $\mathcal{A}_2$ forging signatures is negligible. Hence, our scheme can resist forgery attack.

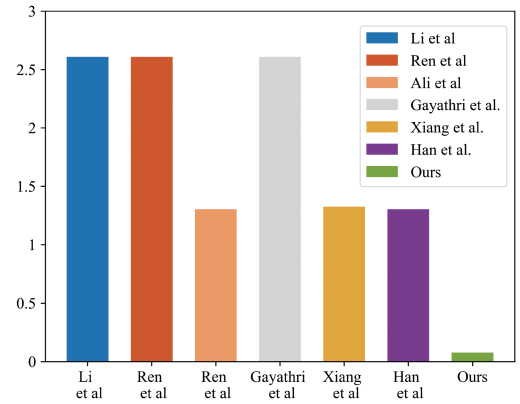

Fig. 2. Comparison of the signature generation cost.

V. Performance Evaluation

In this section, we assess the performance of our scheme by estimating the computational and communication costs, as well as analyzing the delay in writing and querying data within the blockchain.

For ECC-based schemes, we select a nonsingular elliptic curve $\bar{E} : y^2 = x^3 + ax + b \bmod q$, where $a, b \in Z_q^*$. G is an additive group of order q over $\bar{E}$. The size of the G is 40 B and the size of p and q is 20 B.

For pairing-based schemes, we construct the bilinear pairing on 80 b security level, as $\bar{e} : G_1 \times G_1 \rightarrow G_T$, where G_1 is an additive group which is generated on a super singular elliptic curve $\hat{E} : y^2 = x^3 + x \bmod \hat{q}$ with embedding degree 2. The size of the G_1 is 128 B and the size of $\hat{q}$ is 64 B.

A. Computational Costs Comparison

In order to do the comparison of computation overhead, the simulation experiment is performed on a desktop that consists of an Intel(R) Core(TM) i5-1035G1 processor with 1.19 GHZ clock frequency, 8 GB RAM, with the MIRACL library. Table I lists the comparison of the computation overhead between related schemes and our proposed scheme. The several basic operations are defined in the Nomenclature.

As indicated in Table I and Fig. 2, in terms of the process of generating signatures, our scheme exhibits significantly lower computational cost, which is only $2T_{\text{mu}} + 2T_{\text{ha}}$, due to the absence of elliptic curve point multiplication operations. In comparison with other relevant schemes, [28] requires $2T_{\text{Mu}}$, [20] requires $2T_{\text{Mu}}$, [9] requires T_{Mu}, [17] requires $2T_{\text{Mu}}$, and

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TABLE I
COMPARATIVE ANALYSIS OF PERFORMANCE

Scheme	Single-sign (ms)	Single-verify (ms)	Signature size (bytes)
Li [28]	$2T_{\text{Mu}} \approx 2.608$	$2T_{\text{Pa}} \approx 14.332$	$2\|G_1\| = 256$
Ren [20]	$2T_{\text{Mu}} \approx 2.608$	$2T_{\text{Pa}} + 2T_{\text{mu}} \approx 14.355$	$2\|G_1\| = 256$
Ali [9]	$T_{\text{Mu}} \approx 1.304$	$1T_{\text{Pa}} + 1T_{\text{mu}} + 1T_{\text{Mu}} \approx 8.482$	$2\|G_1\| = 256$
Gayathri [17]	$2T_{\text{Mu}} \approx 2.608$	$5T_{\text{Mu}}+3T_{\text{Add}} \approx 6.554$	$\|G + 2Z_q^*\| = 80$
Xiang [18]	$T_{\text{Mu}}+T_{\text{inv}} \approx 1.326$	$4T_{\text{Mu}}+2T_{\text{Add}} \approx 5.239$	$\|G + Z_q^*\| = 60$
Han [14]	$T_{\text{Mu}} \approx 1.304$	$3T_{\text{Mu}}+T_{\text{Add}} \approx 3.923$	$\|G + Z_q^*\| = 60$
Our Scheme	$2T_{\text{mu}}+2T_{\text{ha}} \approx 0.077$	$4T_{\text{Mu}}+3T_{\text{Add}} \approx 5.25$	$\|G + Z_q^*\| = 60$

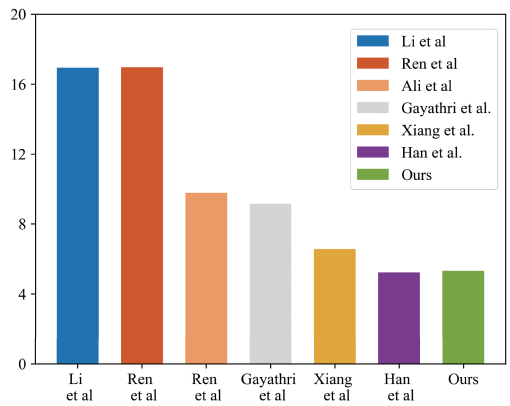

Fig. 3. Comparison of the signature overall cost.

TABLE II
FEATURES COMPARISON

Schemes	Features			
	F1	F2	F3	F4
Li et al.'s [28]	✓	×	✓	✓
Ren et al.'s [20]	✓	×	✓	✓
Ali et al.'s [9]	✓	×	✓	✓
Gayathri et al.'s [17]	✓	×	✓	✓
Xiang et al.'s [18]	×	×	✓	×
Han [14]	✓	×	×	×
Our	✓	✓	✓	✓

F1: conditional identity anonymity,
F2: temporary secret leakage attack
F3: key recover attack, F4: forgery attack

[14] require T_{Mu}, while [18] demands $T_{\text{Mu}} + T_{\text{inv}}$. As a result, our scheme substantially reduces the computational burden on the user side, resulting in a more efficient solution compared to related schemes.

As shown in Table I, for the verification cost, Li et al.'s scheme requires $2T_{\text{Pa}}$, Ren et al.'s scheme requires $2T_{\text{Pa}}$+$2T_{\text{mu}}$, Ali et al.'s scheme requires $1T_{\text{Pa}}$+$1T_{\text{mu}}$+$1T_{\text{Mu}}$, Gayathri et al.'s scheme requires $5T_{\text{Mu}}$+$3T_{\text{Add}}$, Xiang et al.'s scheme requires $4T_{\text{Mu}}$+$2T_{\text{Add}}$, Han et al.'s scheme requires $3T_{\text{Mu}}$+T_{Add}, and our scheme requires $4T_{\text{Mu}} + 3T_{\text{Add}}$. While Han et al.'s scheme has the shortest verification time, it is vulnerable to key recovery attacks as an adversary can collect multiple signatures to decrypt the private key.

According to Table I and Fig. 3, our scheme has an overall signature cost of only 5.327 ms. In comparison, Li et al.'s scheme requires 16.94 ms, Ren et al.'s scheme requires 16.963 ms, Ali et al.'s scheme requires 9.786 ms, Gayathri et al.'s scheme requires 9.162 ms, Xiang et al.'s scheme requires 6.565 ms, and Han et al.'s scheme requires 5.227 ms. It is worth noting that our scheme outperforms Li et al.'s scheme by 68.5%, Ren et al.'s scheme by 68.6%, Ali et al.'s scheme by 45.5%, Gayathri et al.'s scheme by 41.86%, and Xiang et al.'s scheme by 18.55%.

B. Communication Overhead Comparison

The size of the signature plays a crucial role in determining the communication overhead. This is particularly significant in IoT devices, which typically have limited memory bandwidth and operate in low-power consumption modes. It is essential to keep the signature size within an optimal range to prevent excessive communication overhead. From Table I, we can observe that the size of the signature in these scheme [9], [20], [28] are $2|G_1| = 256$ B, the size of the signature in the scheme [17] is $|G + 2Z_q^*| = 80$ B, and the size of the signature in these schemes [14], [18] are $|G + Z_q^*| = 60$ B. Given that our signature length is also 60 B, our approach is suitable for scenarios where communication distance is limited. However, while our scheme's communication cost may not be superior to that of other schemes, it offers enhanced security features, as shown in Table II. This distinguishes our scheme and makes it more robust in terms of protecting sensitive data and ensuring secure communication.

C. Write Latency and Query Latency

This section focuses on investigating the time costs introduced by the adoption of blockchain technology. In our scheme, blockchain write operations involve the authentication server recording authentication information and the PKG appending relevant evidence. Simultaneously, when retracing historical data or when a task publisher retrieves information, it necessitates a blockchain read operation. Therefore, evaluating the performance of the deployed chaincode, particularly the temporal overhead of query and write functions in the smart contract, becomes necessary. On Ubuntu 20.04, we deployed Hyperledger Fabric 2.3, configuring three nodes to simulate consensus servers. The system configuration included Docker 20.10.12, Docker-Compose 1.26.0, and Go 1.17.7. Subsequently, we deployed a smart contract for data writing and querying. Experimental analysis revealed an average time of approximately 2.038 s for a single write operation and about 9.371 ms for a single query. It is noteworthy that the time required for write operations is relatively longer, primarily because the write process must undergo the consensus phase at the nodes to be successfully recorded on the chain. In contrast, query operations only involve reading replica data from the blockchain, resulting in shorter time requirements.

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VI. Conclusion

In this article, security and privacy are critical challenges facing IIoT. To overcome these challenges, we propose a BCCA scheme. The consensus server can efficiently verify batch signatures, and the PKG can dynamically modify the illegal information on the blockchain. Our scheme is also resilient against common attacks. Furthermore, performance analysis indicates relatively low computational costs, demonstrating the feasibility of BCCA for deployment in IIoT. However, we also recognize that in real-world applications, devices from different domains need to collaborate effectively. Therefore, cross-domain identity authentication is an important and unresolved issue. Future research efforts will focus on addressing the security challenges of cross-domain authentication for devices in the IIoT.

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Xinchao Wang received the B.S. degree in cyberspace security from Southwest Petroleum University, Sichuan, China, in 2022. He is currently working toward the M.S. degree in communication and information system with the Nanjing University of Aeronautics and Astronautics, Nanjing, China.

His research interests mainly focus on blockchain technologies, security and privacy in wireless communication.

Wei Wang (Member, IEEE) received the B.Eng. degree in information countermeasure technology and the M.Eng. degree in signal and information processing from Xidian University, Xi'an, China, in 2011 and 2014, respectively, and the Ph.D. degree in electrical and electronic engineering from Nanyang Technological University, Singapore, in 2018.

From 2018 to 2019, he was a Postdoctoral Fellow with the Department of Electrical and Computer Engineering, University of Waterloo, Waterloo, ON, Canada. He is currently a Professor with the Nanjing University of Aeronautics and Astronautics, Nanjing, China. His research interests include wireless communications, space-air-ground integrated networks, wireless security, and blockchain.

Dr. Wang was the recipient of the IEEE Student Travel Grants for IEEE International Conference on Communications 2017, and the Chinese government award for outstanding self-financed students abroad.

Cheng Huang (Member, IEEE) received the B.Eng. and M.S. degrees in information security from Xidian University, Xi'an, China, in 2013 and 2016, respectively, and the Ph.D. degree in electrical and computer engineering from the University of Waterloo, Waterloo, ON, Canada, in 2020.

He is currently a Research Fellow with the Department of Electrical and Computer Engineering, University of Waterloo. His research focuses on security and privacy in vehicular networks, cloud and edge computing, and blockchain technologies.

Dr. Huang was the recipient of multiple awards, including Best Paper Awards at ICCC'15, ICC'18, GLOBECOM'22, and ICCC'23. He is also an Associate Editor for the *Peer-to-Peer Applications and Networking journal* (Springer), and was the Publicity Chair for ICA3PP'22 and PST'23.

Ping Cao received the M.S. degree in electronic and communication engineering from the University of Chinese Academy of Sciences, Beijing, China, in 2014.

He is currently an Engineer with the Nanjing University of Aeronautics and Astronautics, Nanjing, China. His research interests include blockchain and distributed systems.

Youwen Zhu received the B.E. and Ph.D. degrees in computer science from the University of Science and Technology of China, Hefei, China, in 2007 and 2012, respectively.

From 2012 to 2014, he was a JSPS Postdoctoral with Kyushu University, Fukuoka, Japan. He is currently a Professor with the College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, China. He has authored or coauthored more than 40 papers, and was a program committee member in several international conferences. His research interests include information security and data privacy.

Qihui Wu (Senior Member, IEEE) received the B.S. degree in communications engineering and the M.S. and Ph.D. degrees in communications and information systems from the Institute of Communications Engineering, Nanjing, China, in 1994, 1997, and 2000, respectively.

From 2003 to 2005, he was a Postdoctoral Research Associate with Southeast University, Nanjing, China. From 2005 to 2007, he was an Associate Professor with the Institute of Communications Engineering, PLA University of Science and Technology, Nanjing, China, where he is currently a Full Professor. From March 2011 to September 2011, he was an Advanced Visiting Scholar with the Stevens Institute of Technology, Hoboken, NJ, USA. Since 2016, he has been with the Nanjing University of Aeronautics and Astronautics, Nanjing, China, where he has been a Cheung Kong Scholars Distinguished Professor. His current research interests span the areas of wireless communications and statistical signal processing, with emphasis on system design of software defined radio, cognitive radio, and smart radio.

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